

Stochastic gradient descent and robustness to ill-conditioning

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Joint work with Aymeric Dieuleveut, Nicolas Flammarion,
Eric Moulines - Ecole des Ponts, June 2016

“Big data” revolution?

A new scientific context

- **Data everywhere:** size does not (always) matter
- **Science and industry**
- **Size and variety**
- **Learning from examples**
 - n observations in dimension p

Search engines - Advertising

The image shows a screenshot of a web browser displaying a Bing search results page for the query "tour de france". The browser's address bar shows the URL: <https://www.bing.com/search?q=tour+de+france&go=Submit&q=n&form=QBRE&filt=all&pq=tour+de+france&sc=8>. The browser's toolbar includes links to Apps, GMAIL, Intranet, Francis Bach - INRIA, Le Monde, CP, Scholar, Equipe, Agenda, Liberation, and PAMI. Below the toolbar, there are navigation tabs for WEB, IMAGES, VIDEOS, MAPS, NEWS, and MORE, with a "Sign in" link on the right. The search bar contains the text "tour de france" and a magnifying glass icon. Below the search bar, the results are displayed. At the top, it says "121 000 000 RESULTS" and provides filters for "Narrow by language" and "Narrow by region". The first result is "Tour de France 2014" with a link to "Translate this page" and the URL "www.letour.fr". The description for this result is "tour de picardie 2014 - ... ag2r la mondiale; astana pro team; bigmat - auber 93; bmc racing team; bretagne - seche environnement". Below this, there are four more results: "Parcours" (Du samedi 29 juin au dimanche 21 juillet 2013, le 100 e Tour de ...), "Tour de France 2011" (Tour de France 2014 - Site officiel de la célèbre course cycliste Le Tour ...), "Classements" (Classements - Tour de France 2013. Tour de France 2013 - Site officiel ...), and "Étape 14" (Étape 14 - Saint-Pourçain-sur-Sioule > Lyon - Tour de ...). There are also "Nice 2013" (Tour de France 2012 - Site officiel de la célèbre course cycliste Le Tour ...) and "Étape 18" (Étape 18 - Gap > Alpe-d'Huez - Tour de France 2013). On the right side, there is a "Related searches" section with links to "Tracé Tour de France 2014", "Regarder Tour de France Direct", "Classement Général Tour de France", "Itinéraire Tour de France", "Étape Du Tour", "France 2", "Tour de France Cyclisme", and "Tour de France Online". At the bottom, there is a result for "Tour de France 2013" with a link to "Translate this page" and the URL "www.letour.fr/le-tour/2013/fr". The description for this result is "Tour de France 2013 - Site officiel de la célèbre course cycliste Le Tour de France. Contient les itinéraires, coureurs, équipes et les infos des Tours passés." Below this, there is a result for "Tour de France (cyclisme) — Wikipédia" with a link to "Translate this page" and the URL "fr.wikipedia.org/wiki/Tour_de_France_(cyclisme)". The description for this result is "Le Tour de France est une compétition cycliste par étapes créée en 1903 par Henri Desgrange et Géo Lefèvre, chef de la rubrique cyclisme du journal L'Auto. Histoire · Médiatisation du ... · Équipes et participation".

tour de france - Bing

<https://www.bing.com/search?q=tour+de+france&go=Submit&q=n&form=QBRE&filt=all&pq=tour+de+france&sc=8>

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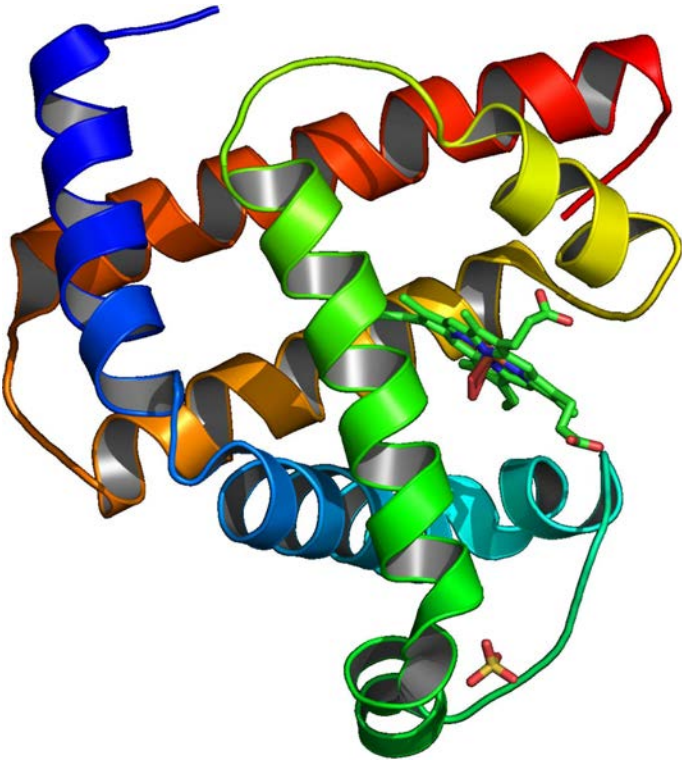
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Related searches
[Tracé Tour de France 2014](#)
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[Classement Général Tour de France](#)
[Itinéraire Tour de France](#)
[Étape Du Tour](#)
[France 2](#)
[Tour de France Cyclisme](#)
[Tour de France Online](#)

Visual object recognition



Bioinformatics



- **Protein:** Crucial elements of cell life
- **Massive data:** 2 millions for humans
- **Complex data**

Context

Machine learning for “big data”

- **Large-scale machine learning:** **large** p , **large** n
 - p : dimension of each observation (input)
 - n : number of observations
- **Examples:** computer vision, bioinformatics, advertising

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 - p : dimension of each observation (input)
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- **Examples:** computer vision, bioinformatics, advertising
- **Ideal running-time complexity:** $O(pn)$
- **Going back to simple methods**
 - Stochastic gradient methods (Robbins and Monro, 1951)
 - Mixing statistics and optimization

Supervised machine learning

- **Data:** n observations $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, $i = 1, \dots, n$, **i.i.d.**
- Prediction as a **linear function** $\langle \theta, \Phi(x) \rangle$ of features $\Phi(x) \in \mathbb{R}^p$
 - Explicit features adapted to inputs (can be learned as well)
 - Using Hilbert spaces for non-linear / non-parametric estimation

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- Prediction as a **linear function** $\langle \theta, \Phi(x) \rangle$ of features $\Phi(x) \in \mathbb{R}^p$
- **(regularized) empirical risk minimization:** find $\hat{\theta}$ solution of

$$\min_{\theta \in \mathbb{R}^p} \quad \frac{1}{n} \sum_{i=1}^n \ell(y_i, \langle \theta, \Phi(x_i) \rangle) \quad + \quad \mu \Omega(\theta)$$

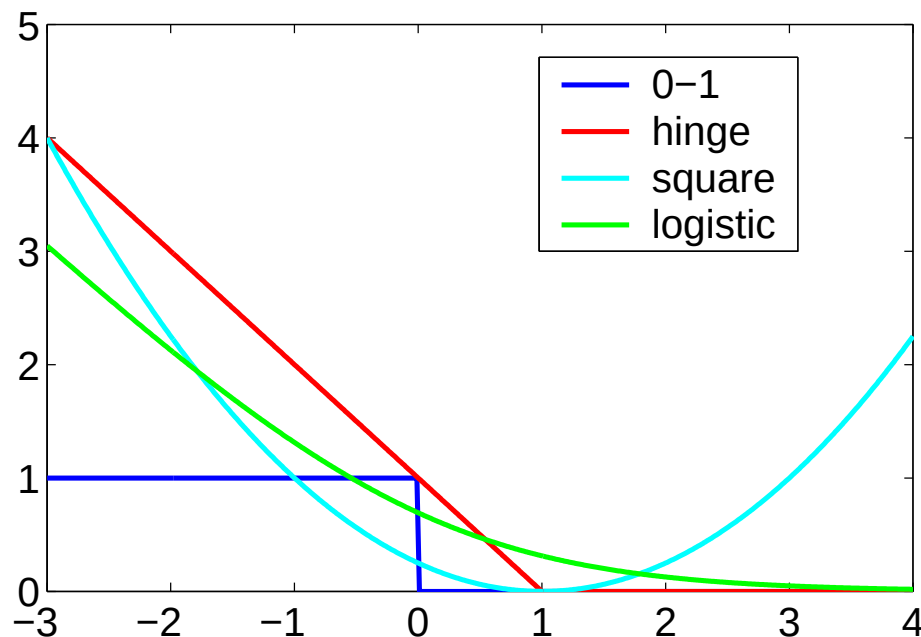
convex data fitting term + regularizer

Usual losses

- **Regression:** $y \in \mathbb{R}$, prediction $\hat{y} = \langle \theta, \Phi(x) \rangle$
 - quadratic loss $\frac{1}{2}(y - \hat{y})^2 = \frac{1}{2}(y - \langle \theta, \Phi(x) \rangle)^2$

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- **Classification :** $y \in \{-1, 1\}$, prediction $\hat{y} = \text{sign}(\langle \theta, \Phi(x) \rangle)$
 - loss of the form $\ell(y \langle \theta, \Phi(x) \rangle)$
 - “True” **0-1** loss: $\ell(y \langle \theta, \Phi(x) \rangle) = 1_{y \langle \theta, \Phi(x) \rangle < 0}$
 - Usual **convex** losses:



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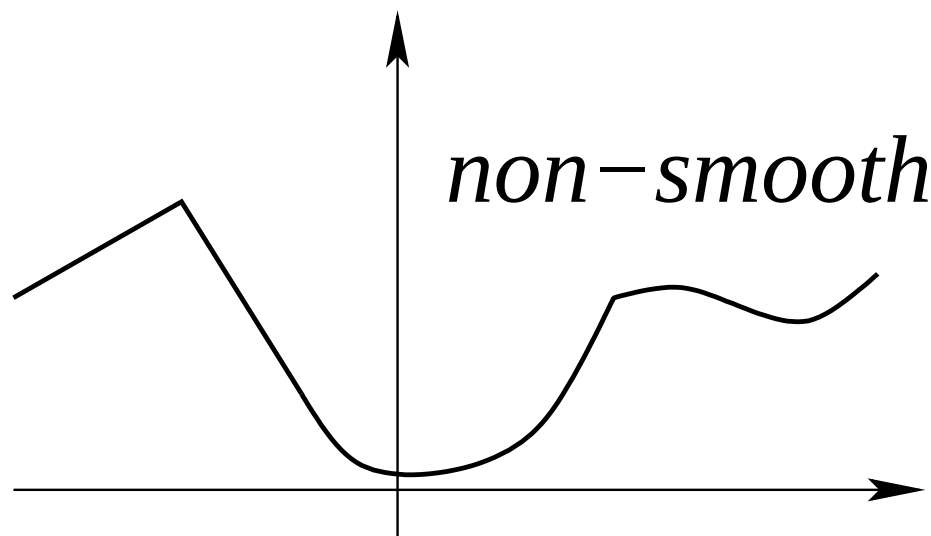
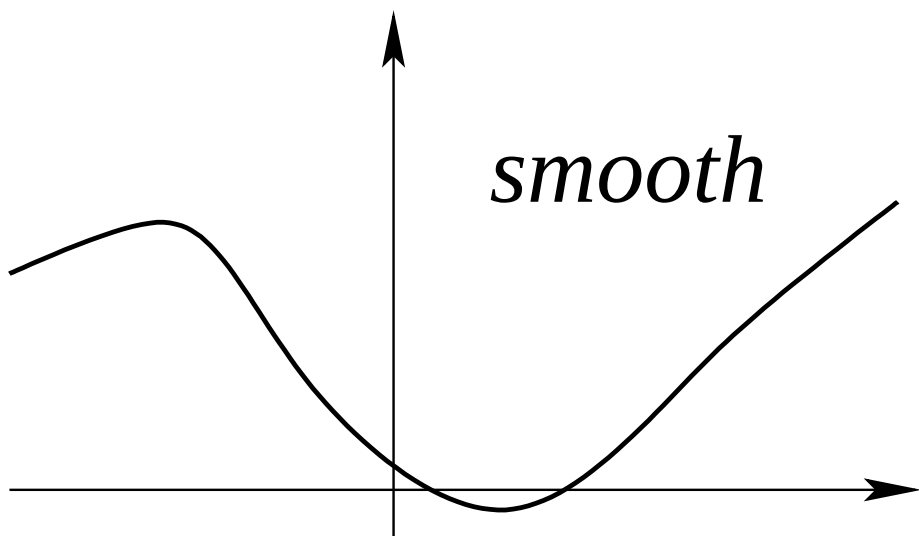
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- Empirical risk: $\hat{f}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, \langle \theta, \Phi(x_i) \rangle)$ **training cost**
- Expected risk: $f(\theta) = \mathbb{E}_{(x,y)} \ell(y, \langle \theta, \Phi(x) \rangle)$ **testing cost**
- **Two fundamental questions:** (1) computing $\hat{\theta}$ and (2) analyzing $\hat{\theta}$

Smoothness and strong convexity

- A function $g : \mathbb{R}^p \rightarrow \mathbb{R}$ is L -smooth if and only if it is twice differentiable and

$$\forall \theta \in \mathbb{R}^p, \text{ eigenvalues}[g''(\theta)] \leq L$$



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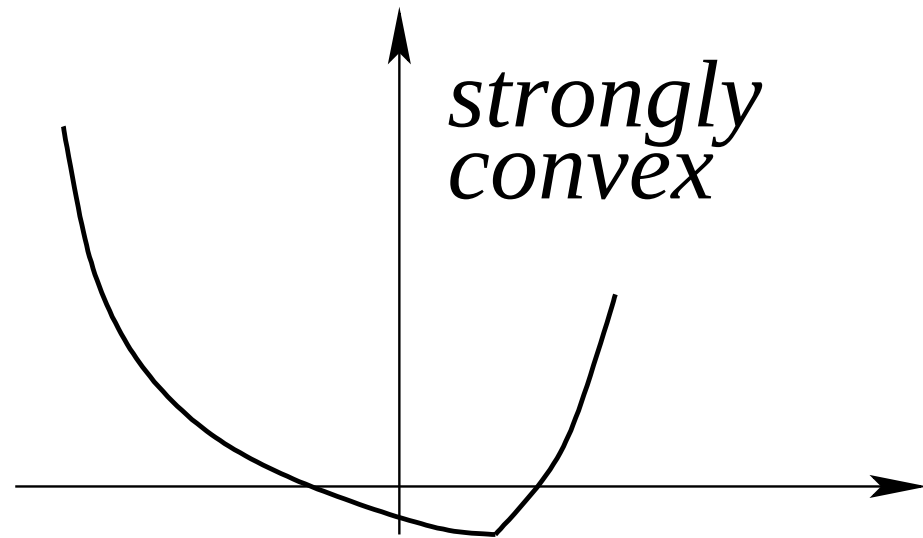
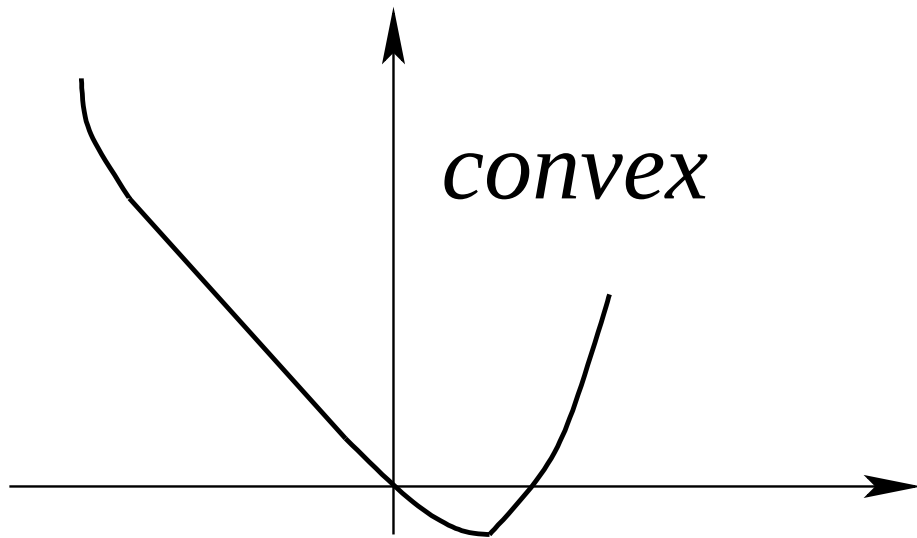
- **Machine learning**

- with $g(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, \langle \theta, \Phi(x_i) \rangle)$
- Hessian $\approx$ covariance matrix $\frac{1}{n} \sum_{i=1}^n \Phi(x_i) \otimes \Phi(x_i)$
- **Bounded data**: $\|\Phi(x)\| \leq R \Rightarrow L = O(R^2)$

Smoothness and strong convexity

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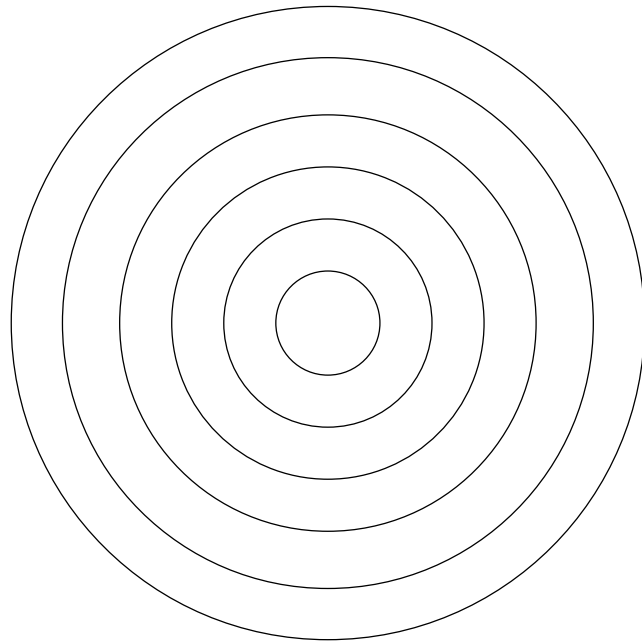
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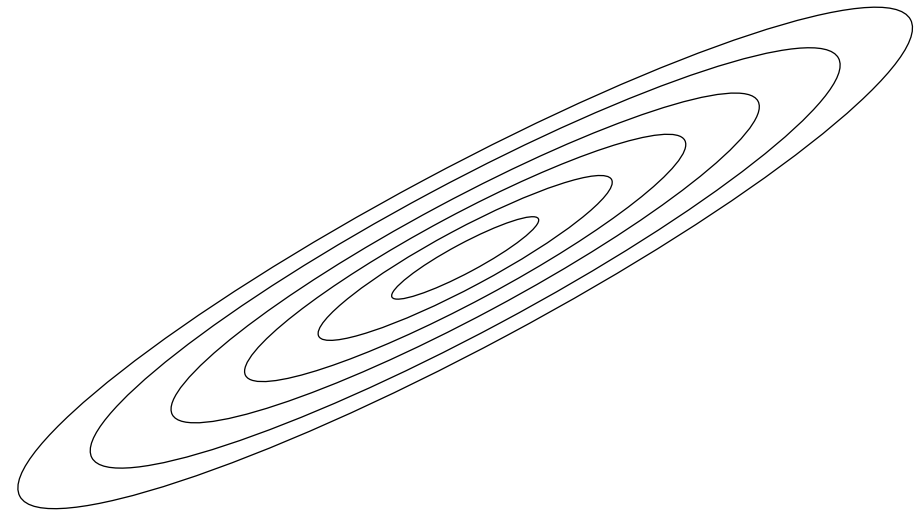
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(large μ/L)



(small μ/L)

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- **Data with invertible covariance matrix** (low correlation/dimension)

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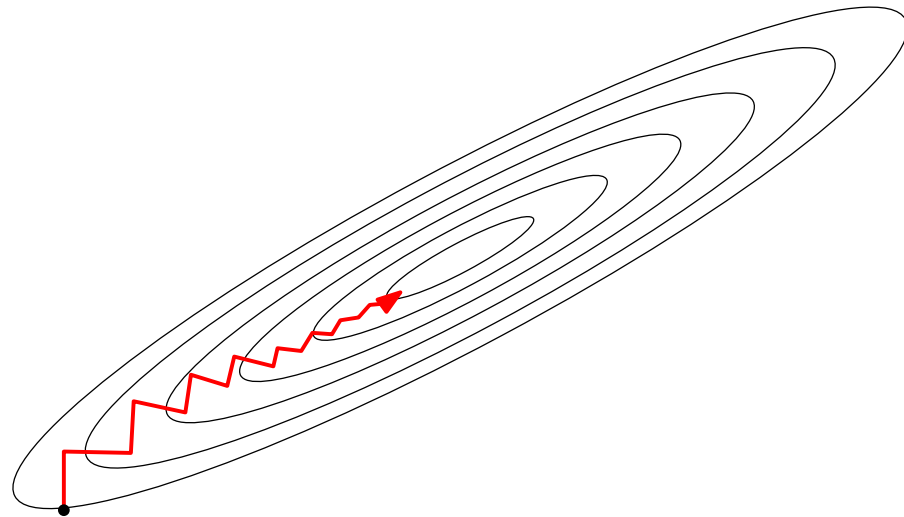
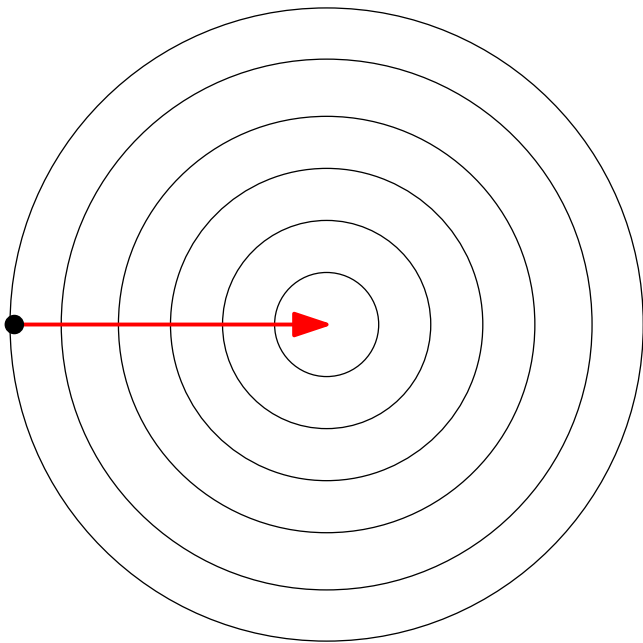
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- **Adding regularization by $\frac{\mu}{2} \|\theta\|^2$**

- creates additional bias unless μ is small

Iterative methods for minimizing smooth functions

- **Assumption:** g convex and smooth on $\mathbb{R}^p$
- **Gradient descent:** $\theta_t = \theta_{t-1} - \gamma_t g'(\theta_{t-1})$
 - $O(1/t)$ convergence rate for convex functions
 - $O(e^{-(\mu/L)t})$ convergence rate for strongly convex functions



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- **Key insights from Bottou and Bousquet (2008)**
 1. In machine learning, no need to optimize below statistical error
 2. In machine learning, cost functions are averages

$\Rightarrow$ **Stochastic approximation**

Stochastic approximation

- **Goal:** Minimizing a function f defined on $\mathbb{R}^p$
 - given only unbiased estimates $f'_n(\theta_n)$ of its gradients $f'(\theta_n)$ at certain points $\theta_n \in \mathbb{R}^p$

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- **Machine learning - statistics**
 - $f(\theta) = \mathbb{E} f_n(\theta) = \mathbb{E} \ell(y_n, \langle \theta, \Phi(x_n) \rangle) =$ **generalization error**
 - **Loss for a single pair of observations:** $f_n(\theta) = \ell(y_n, \langle \theta, \Phi(x_n) \rangle)$
 - Expected gradient:

$$f'(\theta) = \mathbb{E} f'_n(\theta) = \mathbb{E} \{ \ell'(y_n, \langle \theta, \Phi(x_n) \rangle) \Phi(x_n) \}$$

- Beyond convex optimization: see, e.g., Benveniste et al. (2012)

Convex stochastic approximation

- **Key assumption:** smoothness and/or strong convexity
- **Key algorithm:** stochastic gradient descent (a.k.a. Robbins-Monro)

$$\theta_n = \theta_{n-1} - \gamma_n f'_n(\theta_{n-1})$$

– Polyak-Ruppert averaging: $\bar{\theta}_n = \frac{1}{n+1} \sum_{k=0}^n \theta_k$

– Which learning rate sequence γ_n ? Classical setting: $\gamma_n = Cn^{-\alpha}$

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- **Running-time** = $O(np)$
 - Single pass through the data
 - One line of code among many

Convex stochastic approximation

Existing analysis

- Known **global** minimax rates of convergence for **non-smooth** problems (Nemirovsky and Yudin, 1983; Agarwal et al., 2012)
 - **Strongly convex:** $O((\mu n)^{-1})$
Attained by averaged stochastic gradient descent with $\gamma_n \propto (\mu n)^{-1}$
 - **Non-strongly convex:** $O(n^{-1/2})$
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- **Asymptotic analysis of averaging** (Polyak and Juditsky, 1992; Ruppert, 1988)
 - All step sizes $\gamma_n = Cn^{-\alpha}$ with $\alpha \in (1/2, 1)$ lead to $O(n^{-1})$ for **smooth** strongly convex problems

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- **A single algorithm for smooth problems with global convergence rate $O(1/n)$ in all situations?**

Least-mean-square algorithm

- **Least-squares:** $f(\theta) = \frac{1}{2}\mathbb{E}[(y_n - \langle \Phi(x_n), \theta \rangle)^2]$ with $\theta \in \mathbb{R}^p$
 - SGD = least-mean-square algorithm (see, e.g., Macchi, 1995)
 - usually studied without averaging and decreasing step-sizes
 - with strong convexity assumption $\mathbb{E}[\Phi(x_n) \otimes \Phi(x_n)] = H \succcurlyeq \mu \cdot \text{Id}$

Least-mean-square algorithm

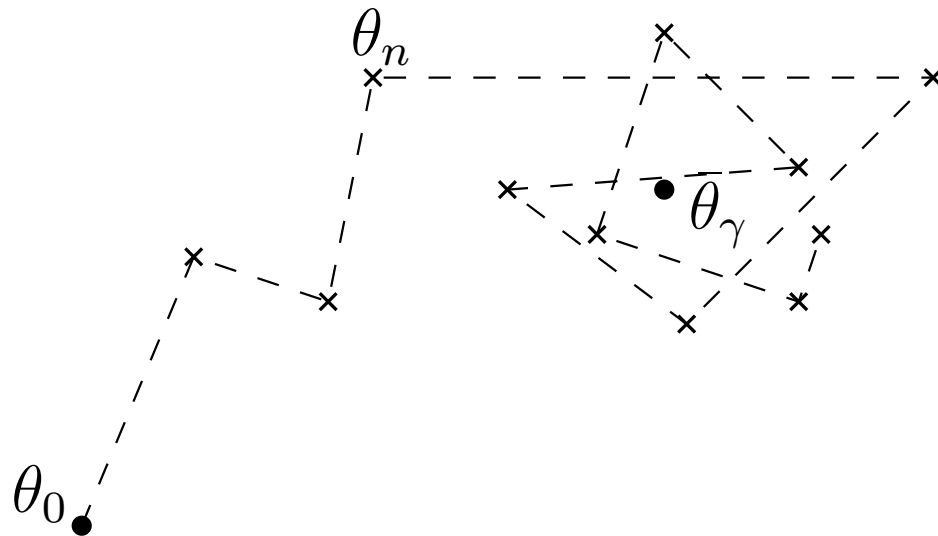
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- **New analysis for averaging and constant step-size** $\gamma = 1/(4R^2)$
 - Assume $\|\Phi(x_n)\| \leq R$ and $|y_n - \langle \Phi(x_n), \theta_* \rangle| \leq \sigma$ almost surely
 - **No assumption regarding lowest eigenvalues of H**
 - Main result:
$$\mathbb{E}f(\bar{\theta}_n) - f(\theta_*) \leq \frac{4\sigma^2 p}{n} + \frac{4R^2 \|\theta_0 - \theta_*\|^2}{n}$$
- **Matches statistical lower bound** (Tsybakov, 2003)
 - Non-asymptotic robust version of Györfi and Walk (1996)

Markov chain interpretation of constant step sizes

- LMS recursion for $f_n(\theta) = \frac{1}{2}(y_n - \langle \Phi(x_n), \theta \rangle)^2$

$$\theta_n = \theta_{n-1} - \gamma(\langle \Phi(x_n), \theta_{n-1} \rangle - y_n)\Phi(x_n)$$

- The sequence $(\theta_n)_n$ is a **homogeneous Markov chain**
 - convergence to a stationary distribution π_γ
 - with expectation $\bar{\theta}_\gamma \stackrel{\text{def}}{=} \int \theta \pi_\gamma(d\theta)$

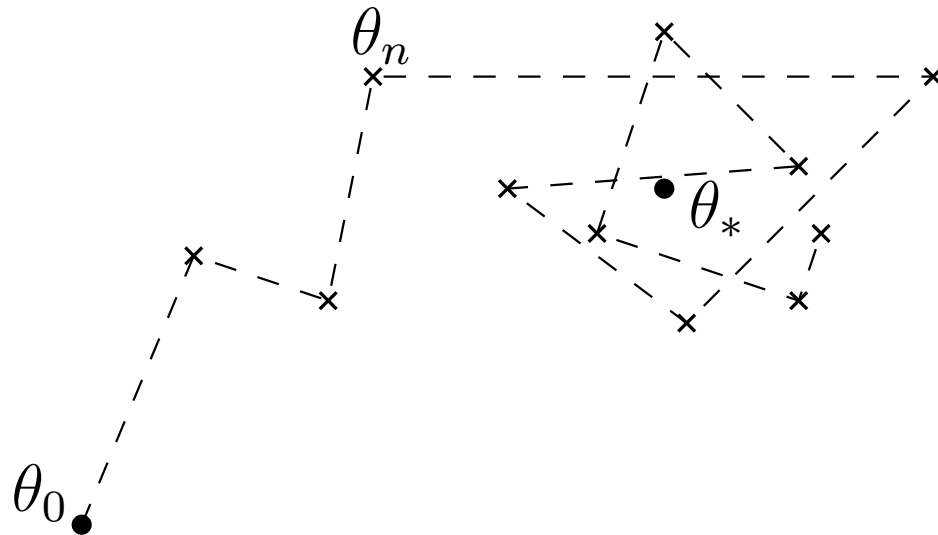


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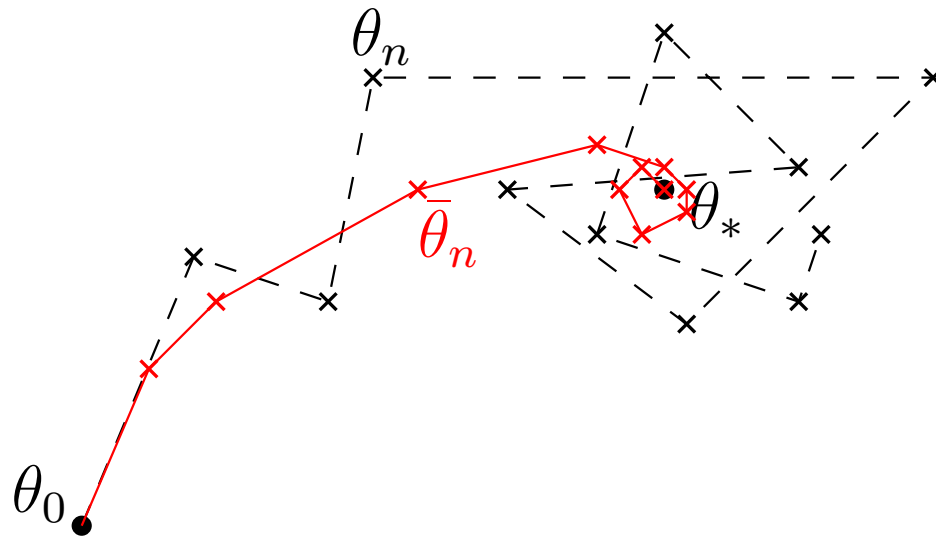


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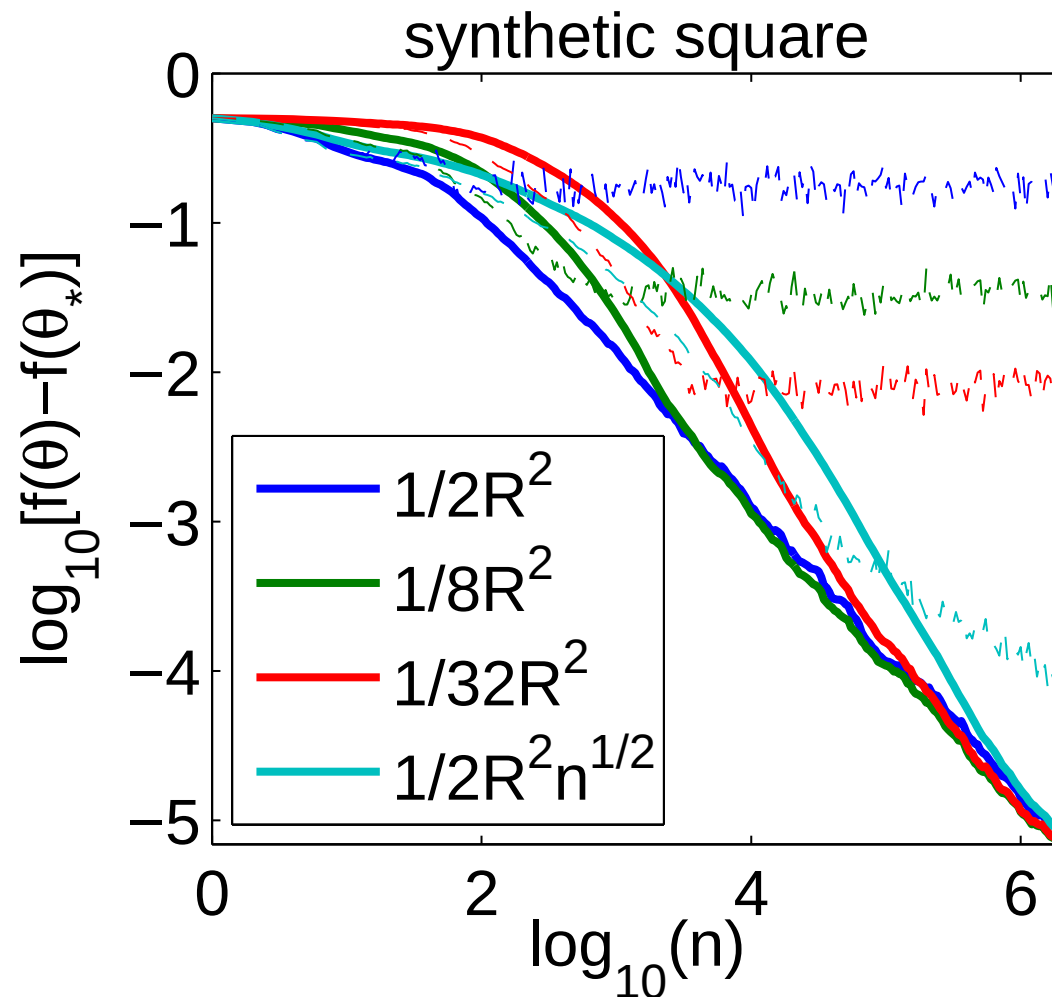
- θ_n does not converge to θ_* but oscillates around it
- oscillations of order $\sqrt{\gamma}$

- **Ergodic theorem:**

- Averaged iterates converge to $\bar{\theta}_\gamma = \theta_*$ at rate $O(1/n)$

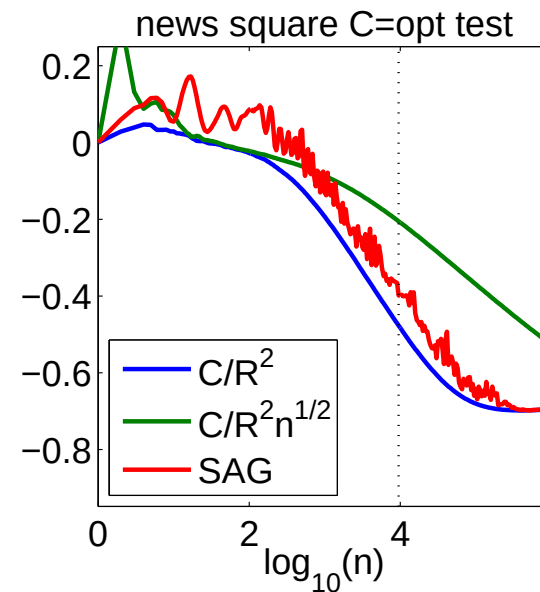
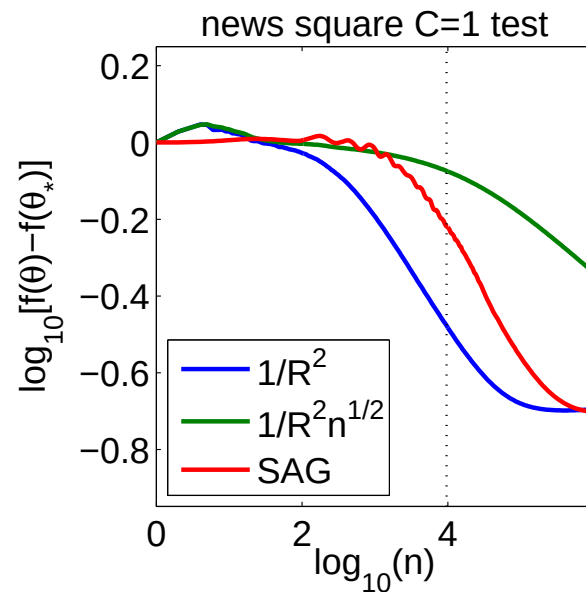
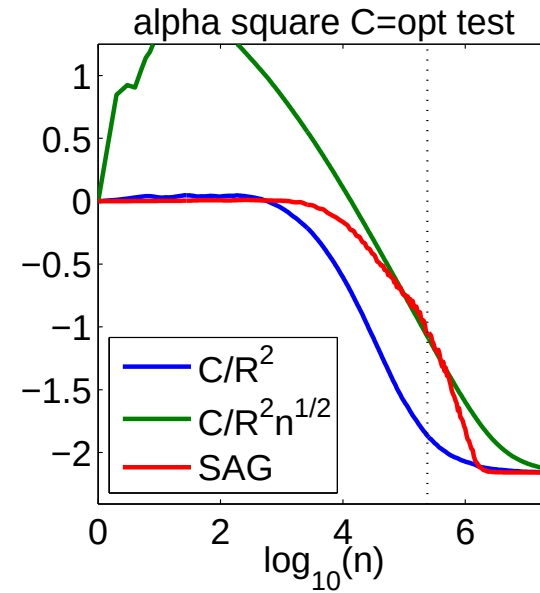
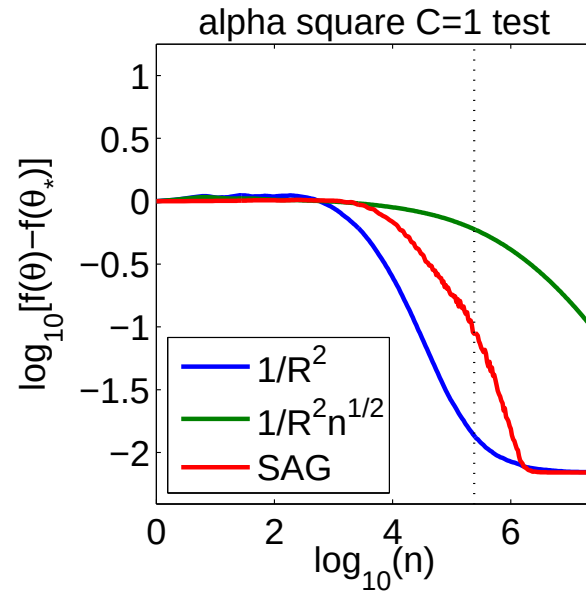
Simulations - synthetic examples

- Gaussian distributions - $p = 20$



Simulations - benchmarks

- *alpha* ($p = 500, n = 500\,000$), *news* ($p = 1\,300\,000, n = 20\,000$)



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- **Least-squares regression**
 - Simpler to analyze and understand
 - **Explicit relationship to bias/variance trade-offs** (next slides)
- **Many important loss functions are not quadratic**
 - **Beyond least-squares with online Newton steps**
 - Complexity of $O(p)$ per iteration with rate $O(p/n)$
 - See Bach and Moulines (2013) for details

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- **Refined assumptions with adaptivity** (Dieuleveut and Bach, 2014)

$$\mathbb{E}f(\bar{\theta}_n) - f(\theta_*) \leq \frac{\sigma^2 \textcolor{blue}{\gamma}^{1/\alpha} \text{tr } \textcolor{blue}{H}^{1/\alpha}}{\textcolor{red}{n}^{1-1/\alpha}} + \frac{\|H^{1/2-r}(\theta_0 - \theta_*)\|_2^2}{\gamma^{2r} \textcolor{red}{n}^{2 \min\{r,1\}}}$$

- SGD is adaptive to the covariance matrix eigenvalue decay
- Leads to optimal rates for non-parametric regression

Achieving optimal bias and variance terms

- Current results with averaged SGD

- **Variance** (starting from optimal θ_*) = $\frac{\sigma^2 p}{n}$

- **Bias** (no noise) = $\min \left\{ \frac{R^2 \|\theta_0 - \theta_*\|^2}{n}, \frac{R^4 \langle \theta_0 - \theta_*, H^{-1}(\theta_0 - \theta_*) \rangle}{n^2} \right\}$

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- **Acceleration is notoriously non-robust to noise** (d'Aspremont, 2008; Schmidt et al., 2011)
 - For non-structured noise, see Lan (2012)

Achieving optimal bias and variance terms

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Achieving optimal bias and variance terms

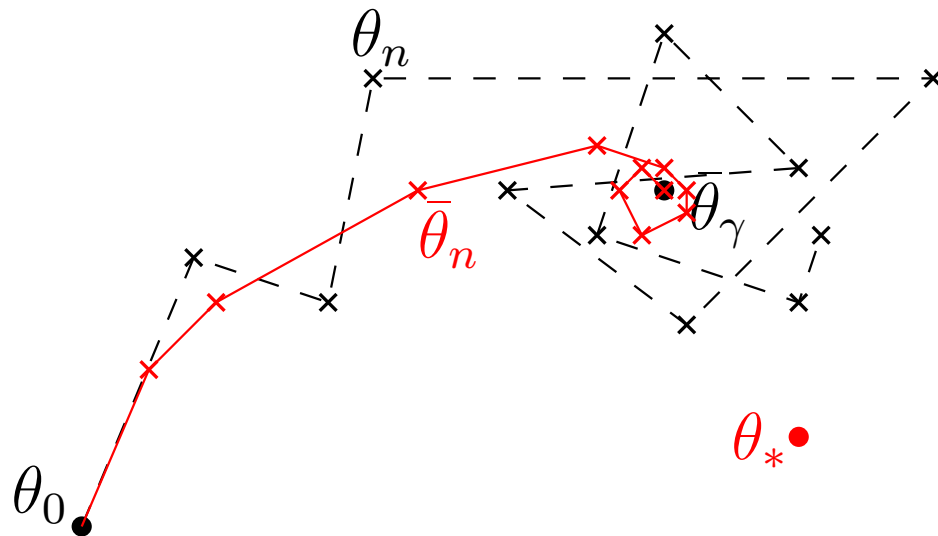
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Beyond least-squares - Markov chain interpretation

- Recursion $\theta_n = \theta_{n-1} - \gamma f'_n(\theta_{n-1})$ also defines a Markov chain
 - Stationary distribution π_γ such that $\int f'(\theta) \pi_\gamma(d\theta) = 0$
 - When f' is not linear, $f'(\int \theta \pi_\gamma(d\theta)) \neq \int f'(\theta) \pi_\gamma(d\theta) = 0$

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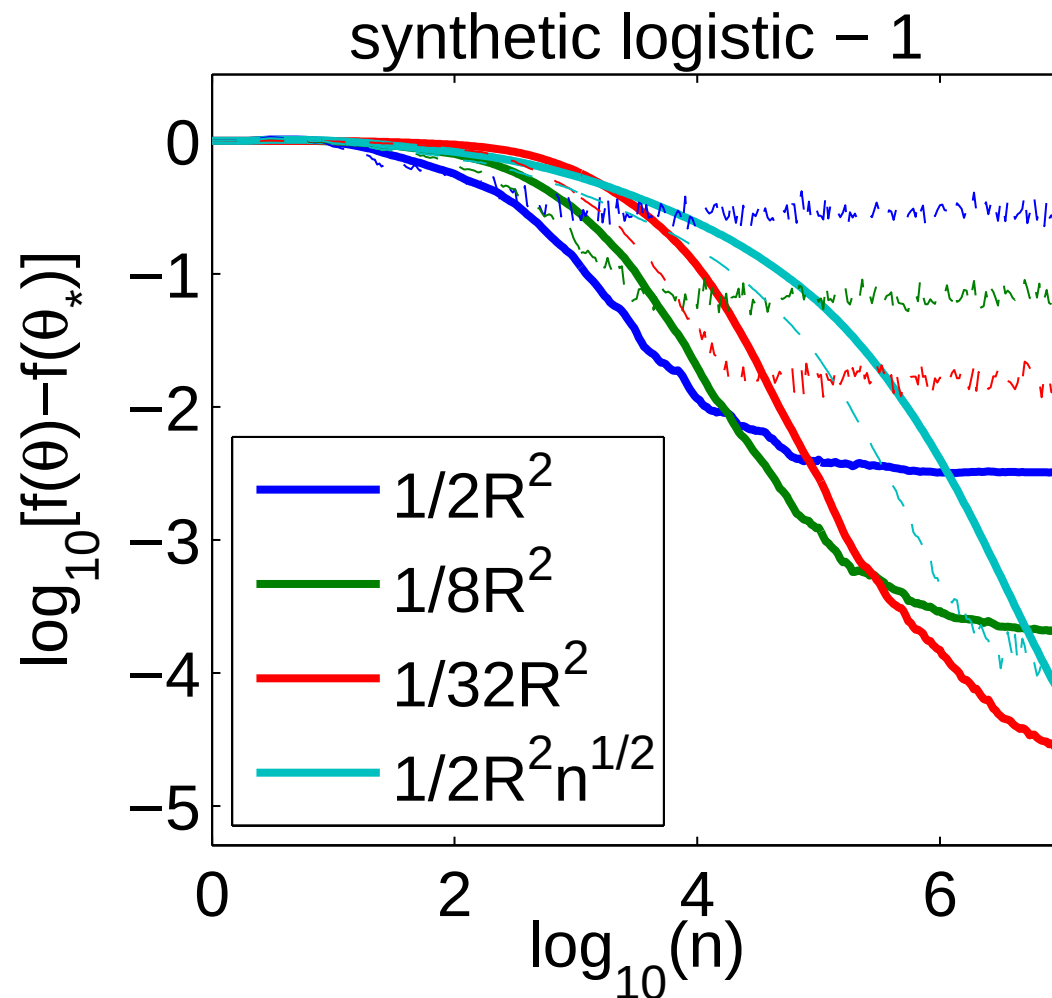


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- θ_n oscillates around the wrong value $\bar{\theta}_\gamma \neq \theta_*$
 - moreover, $\|\theta_* - \theta_n\| = O_p(\sqrt{\gamma})$
- Ergodic theorem
 - averaged iterates converge to $\bar{\theta}_\gamma \neq \theta_*$ at rate $O(1/n)$
 - moreover, $\|\theta_* - \bar{\theta}_\gamma\| = O(\gamma)$ (Bach, 2013)

Simulations - synthetic examples

- Gaussian distributions - $p = 20$



Restoring convergence through online Newton steps

- **Known facts**

1. Averaged SGD with $\gamma_n \propto n^{-1/2}$ leads to *robust* rate $O(n^{-1/2})$ for all convex functions
2. Averaged SGD with γ_n constant leads to *robust* rate $O(n^{-1})$ for all convex *quadratic* functions
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- **Online Newton step**

- Rate: $O((n^{-1/2})^2 + n^{-1}) = O(n^{-1})$
- Complexity: $O(p)$ per iteration

Restoring convergence through online Newton steps

- The Newton step for $f = \mathbb{E}f_n(\theta) \stackrel{\text{def}}{=} \mathbb{E}[\ell(y_n, \langle \theta, \Phi(x_n) \rangle)]$ at $\tilde{\theta}$ is equivalent to minimizing the quadratic approximation

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- **Complexity of least-mean-square recursion for g is $O(p)$**

$$\theta_n = \theta_{n-1} - \gamma [f'_n(\tilde{\theta}) + f''_n(\tilde{\theta})(\theta_{n-1} - \tilde{\theta})]$$

- $f''_n(\tilde{\theta}) = \ell''(y_n, \langle \tilde{\theta}, \Phi(x_n) \rangle) \Phi(x_n) \otimes \Phi(x_n)$ has rank one
- **New online Newton step without computing/inverting Hessians**

Choice of support point for online Newton step

- **Two-stage procedure**

- (1) Run $n/2$ iterations of averaged SGD to obtain $\tilde{\theta}$
- (2) Run $n/2$ iterations of averaged constant step-size LMS
 - Reminiscent of one-step estimators (see, e.g., Van der Vaart, 2000)
 - **Provable convergence rate of $O(p/n)$** for logistic regression
 - Additional assumptions but no **strong convexity**

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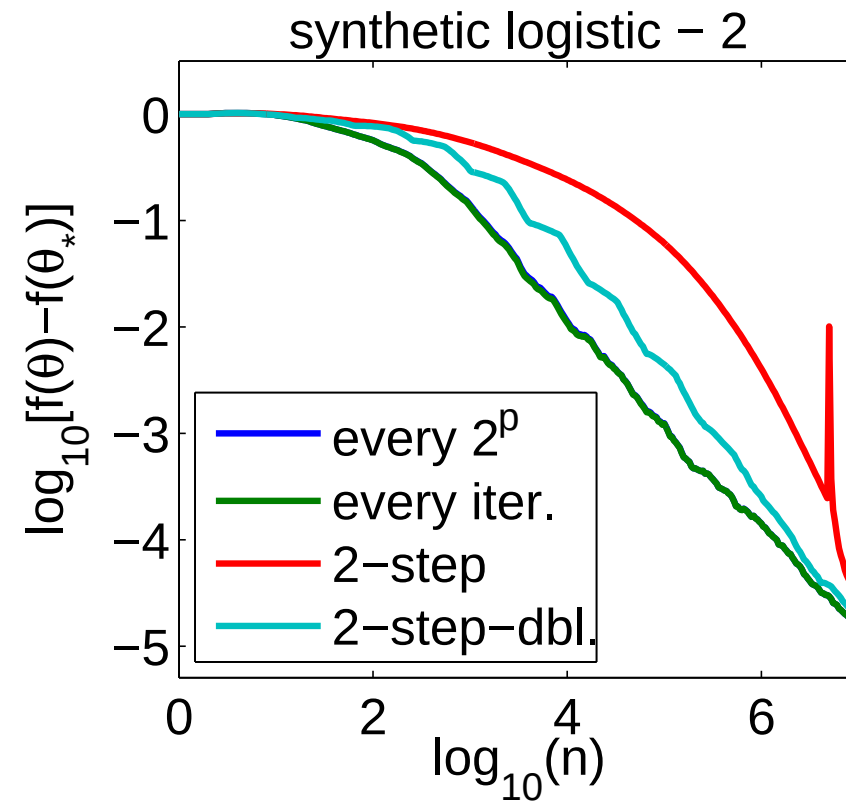
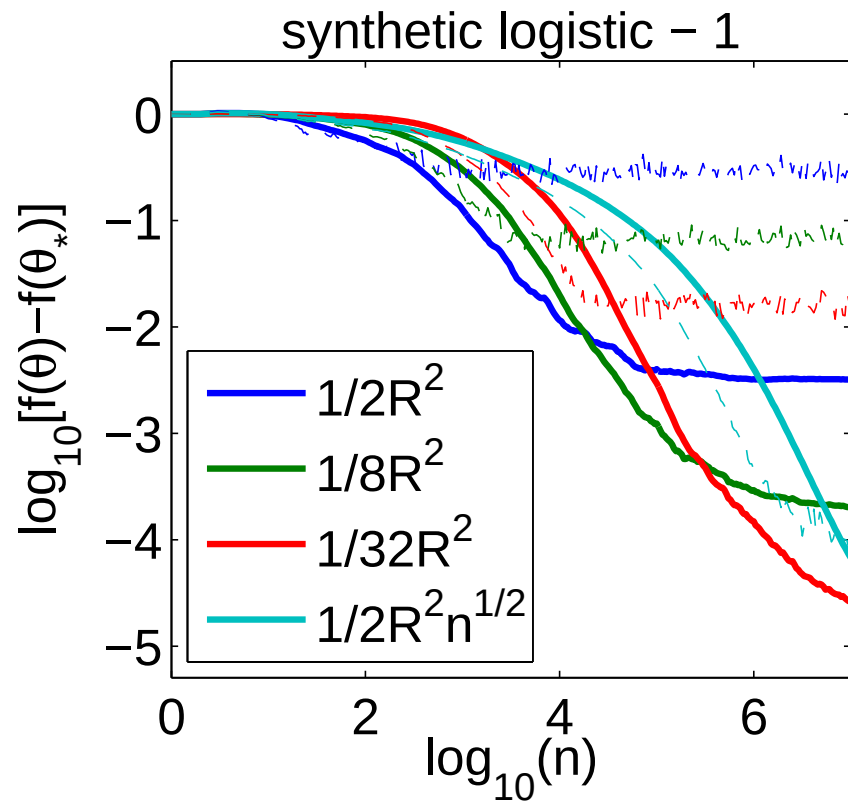
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- **Update at each iteration using the current averaged iterate**

- Recursion:
$$\theta_n = \theta_{n-1} - \gamma [f'_n(\bar{\theta}_{n-1}) + f''_n(\bar{\theta}_{n-1})(\theta_{n-1} - \bar{\theta}_{n-1})]$$
- No provable convergence rate (yet) but best practical behavior
- Note (dis)similarity with regular SGD: $\theta_n = \theta_{n-1} - \gamma f'_n(\theta_{n-1})$

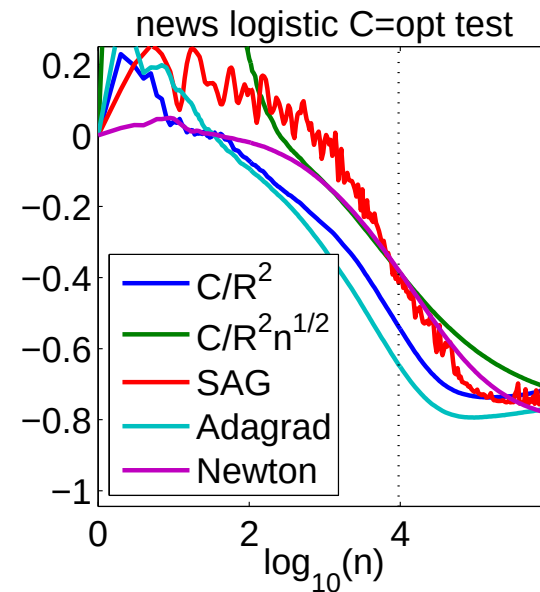
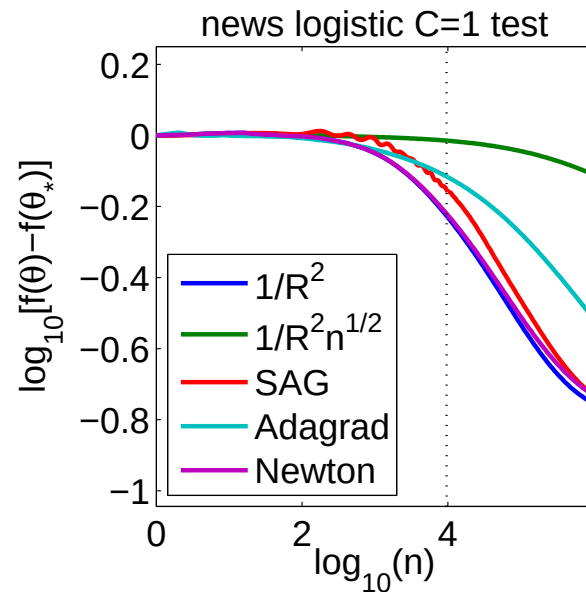
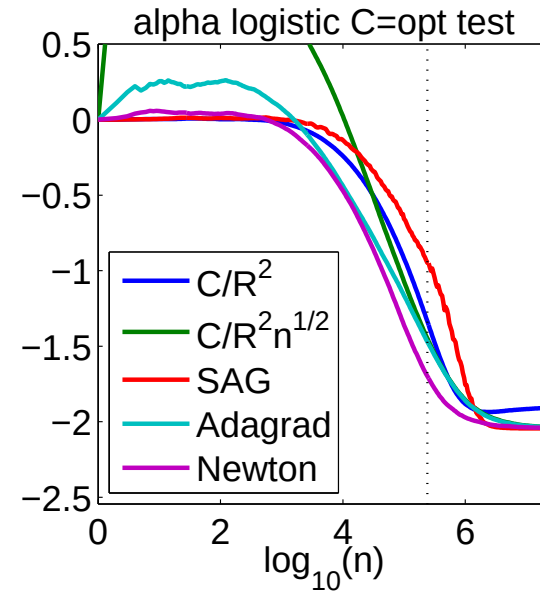
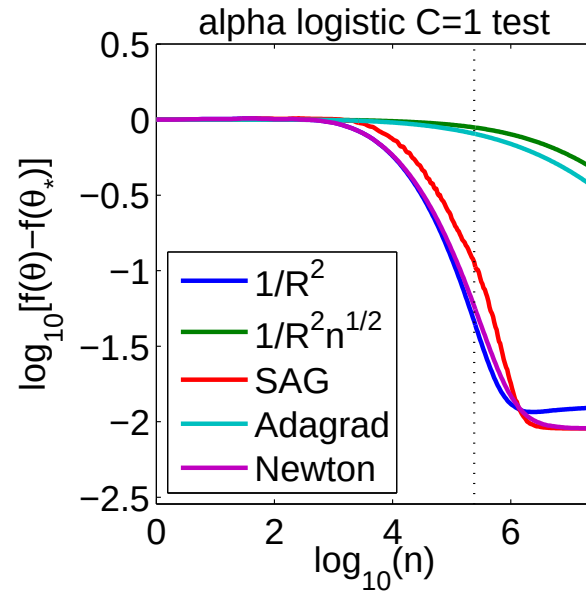
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Conclusions

- **Constant-step-size averaged stochastic gradient descent**
 - Reaches convergence rate $O(1/n)$ in all regimes
 - Improves on the $O(1/\sqrt{n})$ lower-bound of non-smooth problems
 - Efficient online Newton step for non-quadratic problems
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- **Extensions and future work**

- Going beyond a single pass (Le Roux, Schmidt, and Bach, 2012; Defazio, Bach, and Lacoste-Julien, 2014)
- Proximal extensions for non-differentiable terms
- Kernels and nonparametric estimation (Dieuleveut and Bach, 2014)
- **Parallelization**
- **Non-convex problems**

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